10.6 Two Bodies Become One

WORKSHEET

The question

Particles P, of mass 2, and Q, of mass 1, move along a line. Their distances from a fixed point are x_1 and x_2, respectively where $x_2 > x_1$. Each particle is subject to a repulsive force from the other of magnitude $\dfrac{2}{z^3}$, where $z = x_2 - x_1$.

Initially, $x_1 = 0$, $x_2 = 1$, Q is at rest and P moves towards Q with speed 1. Show that z obeys the equation $\dfrac{\mathrm{d}^2z}{\mathrm{d}t^2} = \dfrac{3}{z^3}$.

By first writing $\dfrac{\mathrm{d}^2z}{\mathrm{d}t^2} = v\dfrac{\mathrm{d}v}{\mathrm{d}z}$, where $v = \dfrac{\mathrm{d}z}{\mathrm{d}t}$, show that $z = \sqrt{4t^2 - 2t + 1}$.

By considering the equation satisfied by $2x_1 + x_2$, find x_1 and x_2 in terms of t.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Write down Newton's second law for each particle. In which direction does each force act?

STAGE 1 The separation

Ask yourself Which combination of the two equations involves only z?

STAGE 2 Energy for the separation

Ask yourself With $\ddot{z} = v\frac{\mathrm{d}v}{\mathrm{d}z}$, separate the variables and integrate. What are the initial values of z and $v = \dot{z}$?

STAGE 3 A surprisingly simple solution

Ask yourself The answer to show is $z^2 = 4t^2 - 2t + 1$. So look at $w = z^2$ instead of z. What is $\dot{w}^2$ in terms of w?

STAGE 4 The centre of mass

Ask yourself What equation does $2x_1 + x_2$ satisfy? Why is it the natural combination to look at?

Try this next

1 **Two masses on a string.** (From STEP III, 2023.) Particles A (mass m) and B (mass M) are joined by a light string of length r on a smooth table. Initially A is at the origin and B at $(r, 0)$; B is at rest and A is given velocity u in the y direction. Writing θ for the angle of $\overrightarrow{AB}$ clockwise from the x-axis, show that $\theta = ut/r$, and find $my + MY$ in terms of t. Show that $y = \frac{1}{m+M}\left(mut + Mr\sin\frac{ut}{r}\right)$, and that if $M > m$ the y-velocity of A is sometimes negative.

2 **Attraction.** Repeat the question with an attractive force $\frac{2}{z^3}$. Show that z^2 is still a quadratic in t, and find when the particles collide.

3 **The hidden plane.** For the motion in the question, find the straight line in the plane whose distance from the origin is $z(t)$, and the angle $\theta(t)$ swept out. Check that $z^2\dot{\theta}$ is constant and that $(z^2\dot{\theta})^2 = 3$.

4 Kepler's third law. Two stars of masses m_1 and m_2 move in circles about their centre of mass, a distance a apart, under gravity. Show that the period T satisfies $T^2 = \frac{4\pi^2 a^3}{G(m_1+m_2)}$.

Further reading

- **John Taylor, *Classical Mechanics*** (University Science Books), Chapter 8. The two-body problem, reduced mass and orbits.
- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), Chapter 10, on central forces.
- **Francesco Calogero, *Classical Many-Body Problems Amenable to Exact Treatments*** (Springer). For the ambitious: the world of exactly solvable systems.

The full guided solution, and what the question leads to at university, is in Section 10.6:
https://beyondstep.keithmcnulty.org/ch10/two-body.html